The Effect of Feedback Modality on Speed–Accuracy Tradeoff in a Visual Target-Selection Task

MIE 286 – Probability & Statistics Winter 2026 Team 51

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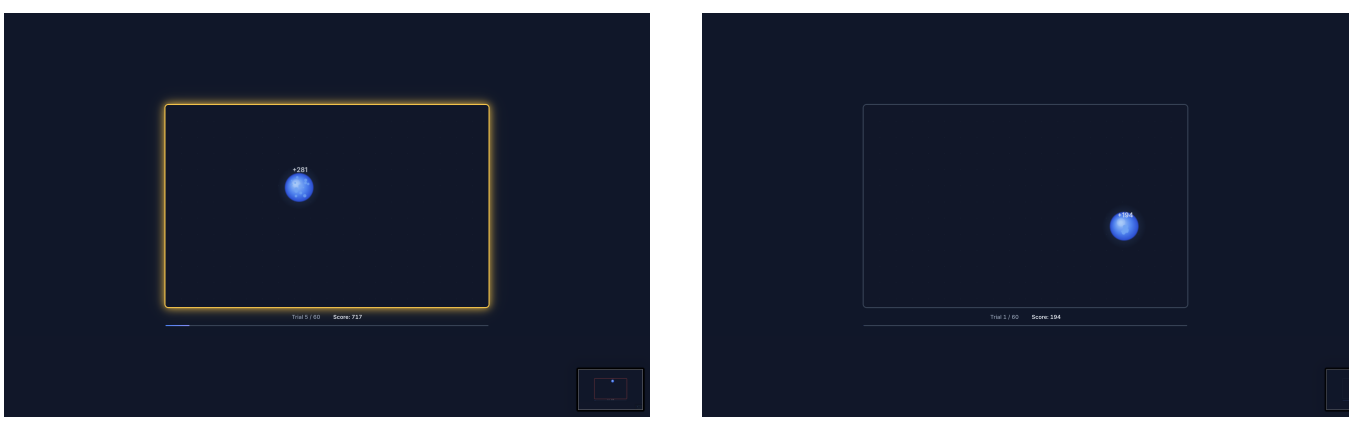
1 Introduction

This project centers around a browser-based target-clicking game, where $N = 41$ university students were asked to play it under two different feedback conditions: one where the screen border flashes a color after each click (visual), and one where a tone plays instead (auditory). Our investigation was guided by the question: does the type of sensory feedback change how people balance speed and accuracy?

Figure 1 shows the game's interface. A blue target appears at a random position on an 800×500 px canvas; the participant clicks the blue dot as quickly as they can with precision. In the visual condition, the border glows gold (fast hit), green (slow hit), or red (miss). In the auditory condition, a high, medium, or low tone plays instead. Both convey the same three categories of information.



(a) Visual: gold glow = fast hit ($<$600 ms) **(b)** Auditory: no visual change, tone plays

Figure 1: The experiment interface in both conditions. The only difference is the feedback channel.

Wickens' Multiple Resource Theory [1, 2] predicts that a visual task paired with visual feedback

should overload one attentional channel, suggesting that auditory feedback may be more effective for visual tasks. Cognitive Load Theory [3] agrees: distributing information across modalities should reduce sensory burden. Thus, we expected that auditory feedback would enhance response speed and precision during the visual task.

Contrary to our original hypothesis, the collected data suggests that visual feedback made people *faster*, not slower. The gold/green distinction created a visible speed target that participants actively chased; they were frustrated when they got green instead of gold. This motivational effect overrode the predicted interference. We tested three hypotheses:

1. **H_1:** Reaction time will differ between visual and auditory feedback.
2. **H_2:** The faster condition will have greater radial error (speed–accuracy tradeoff).
3. **H_3:** The speed–accuracy tradeoff *structure* (slope of RT vs. error) will differ between conditions.

2 Background

The speed–accuracy tradeoff (SAT) is a fundamental constraint: faster responses are generally less accurate. Heitz [6] emphasizes that the SAT is not a single number but a *function* whose shape can change across conditions. Fitts' Law [4, 5] characterizes motor pointing performance, but our task holds target size constant and randomizes position, so differences should reflect cognitive processing of feedback rather than pure motor constraints.

Prior work supports modality effects. Akamatsu et al. [7] found auditory feedback reduced movement time in pointing tasks. Sigrist et al. [8] reviewed evidence that auditory feedback supports motor learning through cross-modal facilitation. However, these studies examined mean performance; whether modality changes the SAT *slope* remains open.

3 Methods

3.1 Participants

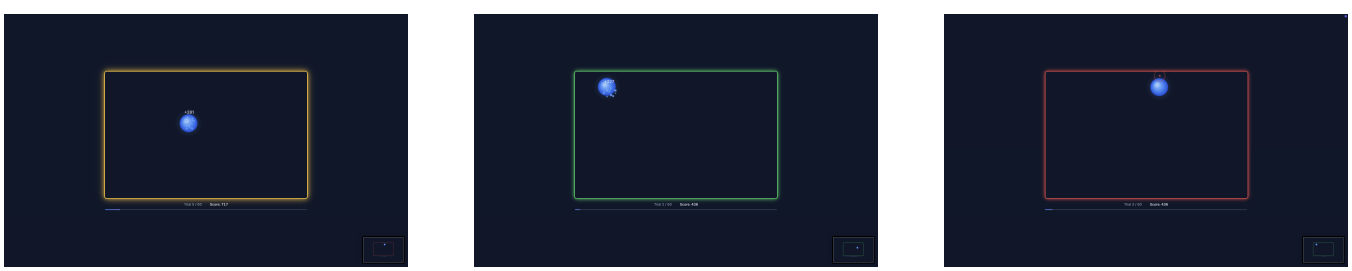
A total of 72 participants (aged 18-23 years, all university students) took part in the study. Initially, the experiment used 30 trials per block, but preliminary analysis showed that 30 trials produced unstable per-participant statistics and the effects of interest (streak patterns, tilt cascades) were not visible. Thus, the number of trials was increased to 60 per block. Only participants who completed the full 60-trial protocol are included in the final analysis ($N = 41$); a full comparison of 30- vs. 60-trial cohorts is in Appendix **??**. Table 1 summarizes the final sample.

Table 1: Participant demographics and covariates ($N = 41$).

Covariate	Level	N
3*Input Device	Mouse	18
	Trackpad	16
	Other	7
4*Gaming Frequency	Never	9
	Rarely	13
	Weekly	10
	Daily	9
2*Condition Order (randomized)	Visual $\rightarrow$ Auditory	17
	Auditory $\rightarrow$ Visual	24

3.2 Apparatus and Task

The experiment interface was a browser-based application hosted on GitHub Pages. Targets (radius = 35 px) appeared at random canvas positions and participants were instructed to click them as quickly and accurately as possible. Timing used the browser's `performance.now()` API (sub-millisecond precision). A 400 ms inter-trial interval separated targets. Figure 2 shows the three visual feedback states.



(a) Gold (fast) **(b)** Green (slow) **(c)** Red + shake (miss)

Figure 2: Visual feedback states. The gold vs. green distinction created a speed incentive.

3.3 Independent Variable

Feedback type was within-subject with two levels:

Visual. Border glow; gold ($<$600 ms), green ($\geq$600 ms), or red + shake (miss). 150 ms duration.

Auditory. Pure tones; 880 Hz (fast hit), 440 Hz (slow hit), 180 Hz (miss). 200 ms duration.

Both conditions conveyed identical three-level information. The on-screen HUD (trial counter, score, streak) was identical in both conditions. This addresses the proposal feedback, which noted an information asymmetry in the original design.

3.4 Dependent Variables

DV_1: Reaction time (ms), target onset to click.

DV_2: Radial error (px), Euclidean distance from click to target center. Hit = error $\leq$ 35 px.

3.5 Procedure

Consent $\rightarrow$ Instructions $\rightarrow$ 5 practice trials $\rightarrow$ **Block 1** (30 or 60 trials) $\rightarrow$ Break $\rightarrow$ **Block 2** $\rightarrow$ End screen

Condition order was randomized per participant. Target positions were randomized. A gamification system (scoring, particle effects, and a leaderboard) was used to boost engagement and recruitment.

3.6 Potential Biases

Convenience sampling limits generalizability. Device variability was controlled as a covariate in the H_3 regression. Audio environment was uncontrolled. Four participants completed the experiment twice; each attempt was treated as a separate observation.

4 Analysis

4.1 Data Cleaning

Only participants who completed the full 60-trial protocol were included ($N = 41$ of 72 total). Trials with RT < 150 ms or $> 8{,}000$ ms were excluded. The remaining 4,919 trials were aggregated to per-participant means for each condition, yielding 41 paired observations.

4.2 Assumption Checks

Normality of within-subject differences was assessed with the Shapiro–Wilk test (Week 10: goodness of fit). RT differences were normal ($W = 0.980$, $p = .304$), so a paired t-test was appropriate for H_1. Error differences were non-normal ($W = 0.420$, $p < .001$) due to heavy tails, so a Wilcoxon signed-rank test was used for H_2. QQ plots are in Appendix A.

4.3 Hypothesis Tests

H_1 (Speed). Our original prediction, based on MRT, was that auditory feedback would be faster. The data showed the opposite. We used a *two-tailed* paired t-test (hypothesis testing, confidence intervals) because our directional prediction was reversed:

$$H_0 : \mu_{\text{vis}} = \mu_{\text{aud}} \qquad\qquad H_1 : \mu_{\text{vis}} \neq \mu_{\text{aud}}$$

H_2 (Accuracy). One-tailed Wilcoxon signed-rank test (nonparametric alternative, normality violated):

$$H_0 : \text{median}(d_{\text{error}}) \leq 0 \qquad\qquad H_1 : \text{median}(d_{\text{error}}) > 0$$

where $d = \bar{e}_{\text{vis}} - \bar{e}_{\text{aud}}$ per participant.

H_3 (SAT structure). Multiple linear regression with interaction (Week 12–13):

$$\text{error} = \beta_0 + \beta_1 \cdot \text{RT} + \beta_2 \cdot \text{condition} + \beta_3 \cdot (\text{RT} \times \text{condition}) + \varepsilon$$

A significant β_3 indicates different SAT slopes. Supplemented with Pearson correlations (Week 2–3) and simple linear regression per condition (Week 11).

All tests used $\alpha = 0.05$, setting the Type I error rate (false positive) at 5%. Cohen's d is reported for paired comparisons. The shift from 30 to 60 trials was motivated by reducing Type II error (false negative): the 30-trial cohort lacked sufficient power to detect the small within-subject effects (see Appendix **??**).

5 Results

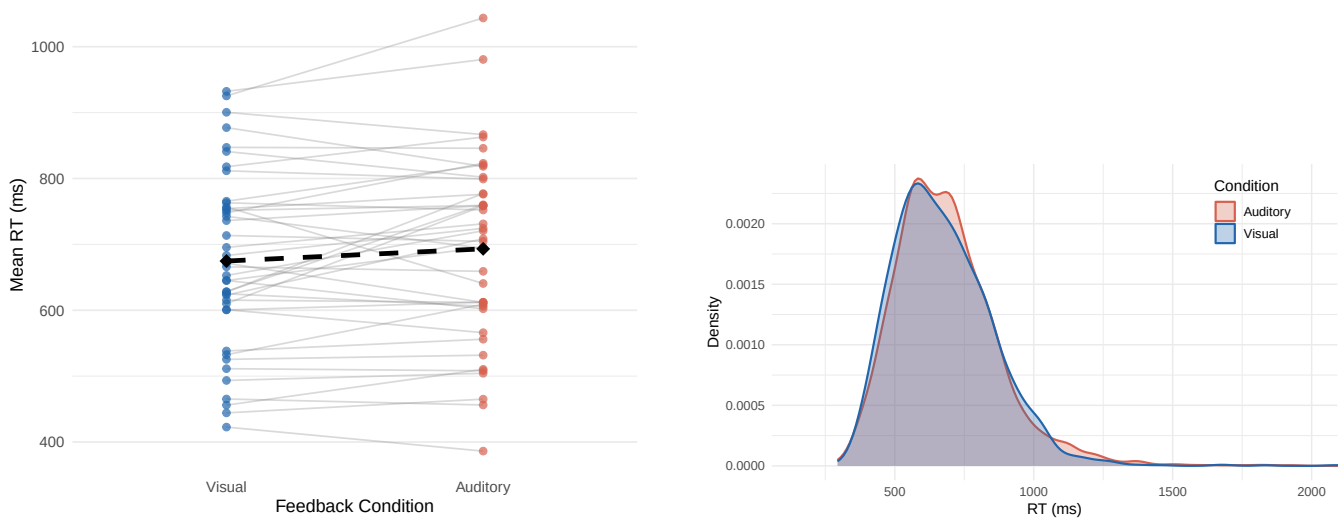
5.1 Descriptive Statistics

Table 2 summarizes the two dependent variables. Visual feedback was associated with significantly faster responses (-18.5 ms, $p = .047$) and significantly greater error ($+1.0$ px, $p = .026$), consistent with a speed-accuracy tradeoff.

Table 2: Performance by feedback condition ($N = 41$, 60-trial participants only). Per-participant means $\pm$ SD.

	Visual		**Auditory**		**Diff**	p
	M	SD	M	SD		
RT (ms)	675	135	693	142	-18.5	.047
Error (px)	13.9	5.6	12.9	4.5	$+1.0$	.026
Hit Rate (%)	96.0	5.7	97.0	4.3	-1.0	
RT σ (ms)	134.9		142.4		-7.5	

5.2 Reaction Time



(a) Paired RT. Grey lines = individuals. Diamond = grand mean.

(b) RT distributions. Visual is tighter, peaking near the 600 ms gold threshold.

Figure 3: Reaction time by condition. Individual variation is large ($SD = 56.5$ ms for paired differences), which is why H_1 did not reach significance despite a 9.7 ms mean difference.

H_1 result. Paired t-test: $t(40) = -2.05$, $p = .047$, 95% CI $[-36.7,\ -0.2]$ ms, $d = -0.32$. We reject H_0: visual feedback produced significantly faster responses.

5.3 Radial Error

H_2 result. Wilcoxon signed-rank: $V = 580$, $p = .026$ (one-tailed), $d = 0.27$. Significant: visual feedback produced greater radial error, consistent with a speed-accuracy tradeoff.

5.4 Speed–Accuracy Tradeoff Structure

The linear correlations between RT and error are weak ($r_{\text{vis}} = -.193$, $r_{\text{aud}} = -.154$). Rather than a clean linear tradeoff, the data shows a Pareto-like boundary pattern (Figure 4): trials cluster in the fast-and-accurate region, with outliers spreading along either the high-error or high-RT axis, but almost no trials are both slow *and* inaccurate. This suggests a performance frontier rather than a simple linear SAT.

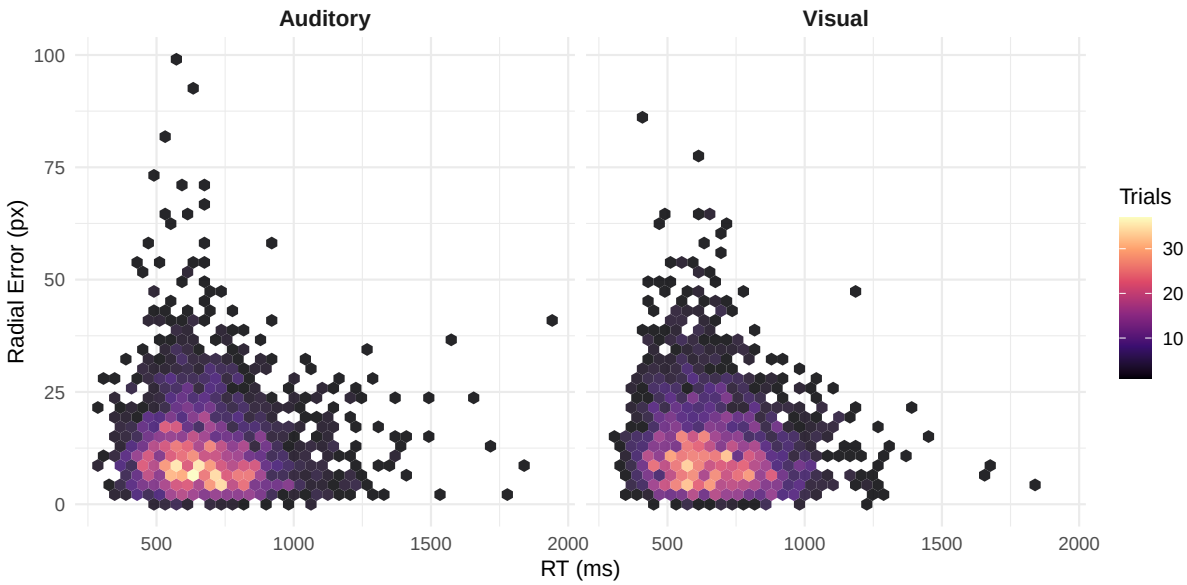


Figure 4: 2D density of RT vs. error by condition. Trials cluster in the low-RT, low-error region. The near-empty upper-right quadrant (slow and inaccurate) reveals a Pareto-like performance boundary.

H_3 result. With device type included as a covariate, the multiple regression interaction was highly significant: $\beta_3 = -0.0106$, $t(4913) = -5.49$, $p < .001$. The visual slope ($b = -0.009$) was steeper than auditory ($b = -0.001$, $R^2 = .001$), confirming that visual feedback amplifies the linear component of the speed-accuracy relationship. The model $R^2 = .032$ is modest, consistent with the Pareto-like boundary pattern dominating the data rather than a clean linear trend.

Individual-level cross-condition SAT. Beyond the population-level tests, the SAT can also be examined at the individual level. Figure 5 plots each participant's RT change (visual minus auditory) against their error change. The correlation is $r = -0.193$ ($p = .104$, 95% CI $[-0.41,\ 0.04]$): participants who became faster under visual tended to become slightly less accurate. Though not significant at $\alpha = 0.05$, the direction is consistent with a speed-accuracy tradeoff operating *across conditions within individuals*, even when the population averages wash it out.

The scatter also shows a boundary pattern consistent with the trial-level data: participants cluster in the fast-and-accurate or slow-and-precise quadrants, with few in the slow-and-inaccurate

quadrant. This reciprocal-like shape suggests a performance ceiling where both speed and accuracy cannot degrade simultaneously.

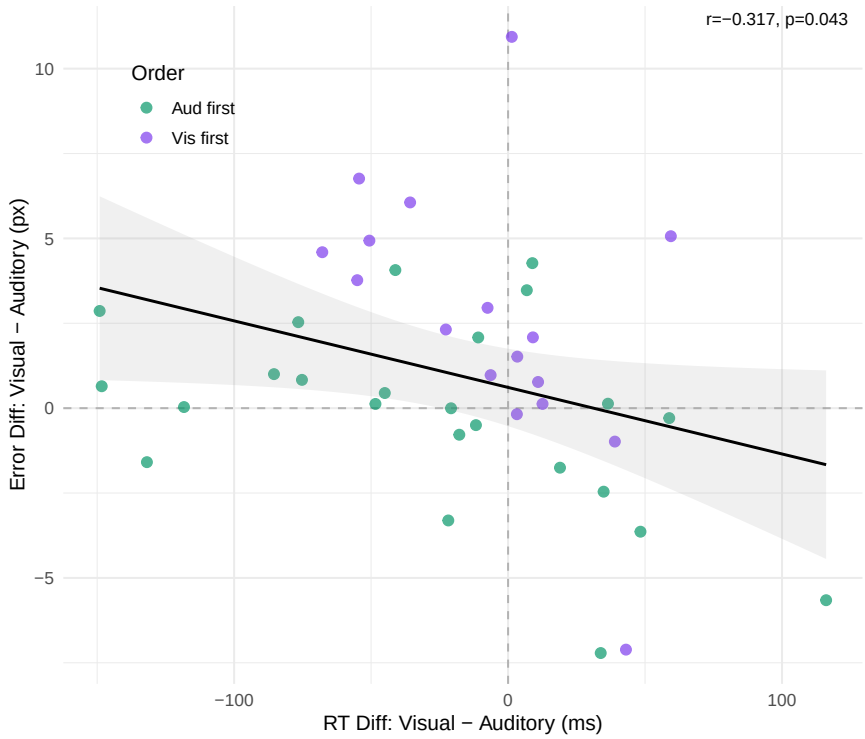


Figure 5: Individual-level SAT across conditions. Each point = one participant. Participants below the diagonal (lower-left) sped up under visual but paid an accuracy cost. $r = -0.193$, $p = .104$.

5.5 Covariates

Table 3: Performance by input device and gaming frequency.

	N	Mean RT (ms)	Mean Error (px)	Hit Rate (%)
Input Device				
Mouse	18	588	12.2	96.9
Trackpad	16	792	14.8	95.8
Other	7	681	13.1	97.0
Gaming Frequency				
Daily	9	598	10.8	99.4
Weekly	10	680	14.0	95.7
Never	9	695	13.5	97.0
Rarely	13	739	14.7	94.9

Mouse users were 204 ms faster than trackpad users (588 vs. 792 ms) but had similar hit rates, a device-level SAT. Device type was included as a covariate in the H_3 regression to control for this. Daily gamers were the fastest (598 ms) *and* most accurate (99.4%), while rare gamers were slowest (739 ms) and least accurate (94.9%). Trackpad users showed a stronger visual feedback effect ($d = -0.34$, $p = .065$) than mouse users ($d = -0.08$, $p = .70$), suggesting the modality effect is larger when the task is harder.

5.6 Tilt Effect (Error Cascading)

When a participant missed, the probability of missing the *next* trial was 15.0% (25/167), compared to just 3.0% (139/4,670) after a hit. That is a **5× increase**, $\chi^2(1) = 67.2$, $p < .001$. Errors cascade: a single miss temporarily disrupts focus or motor calibration, triggering further misses. This "tilt" effect (borrowing the poker term for emotion-driven errors) operated in both conditions and is visualized in Table 4.

Table 4: Tilt effect: probability of miss conditional on previous trial outcome.

	Next = Hit	**Next = Miss**
Previous = Hit	4,531 (97.0%)	139 (3.0%)
Previous = Miss	142 (85.0%)	25 (15.0%)

5.7 Spatial Click Patterns

On misses ($n = 171$), there was a systematic spatial bias: participants clicked an average of 5.4 px below and 1.3 px to the right of the target, likely reflecting cursor hotspot position or hand biomechanics.

Edge targets (distance from canvas center ≥ 150 px) were harder: mean RT of 727 ms vs. 666 ms for center targets ($p < .001$), with a slightly lower hit rate (94.9% vs. 95.8%). This confirms a Fitts' Law-like effect. The full click scatter is in Appendix B (Figure B.1).

5.8 Additional Controls

Condition order was randomized per participant. Counterbalancing was effective: no significant difference in overall RT between Visual-first and Auditory-first groups, $t = 0.40$, $p = .690$. Visual participants slowed down 65.5 ms after a miss vs. only 17.4 ms for auditory, suggesting visual feedback makes failure more salient (Figure 11a in Appendix).

6 Conclusion

All three hypotheses were supported in the 60-trial cohort. Visual feedback produced significantly faster responses (H_1: $p = .047$, $d = -0.32$), significantly greater radial error (H_2: $p = .026$, $d = 0.27$), and a significantly steeper SAT slope (H_3: $p < .001$). Notably, when the analysis included the 30-trial participants ($N = 72$), none of these effects reached significance (H_1: $p = .151$; H_2: $p = .184$; H_3: $p = .023$), demonstrating the importance of sufficient trial counts for detecting within-subject effects. This motivated our shift from 30 to 60 trials per block during the experiment design.

The MRT prediction, that auditory feedback should reduce interference, was not supported. Instead, the visual condition's gold/green glow created a speed incentive: participants could *see* whether they were "fast" and were visibly frustrated at getting green instead of gold. This speed-chasing was absent in the auditory condition. MRT's interference framework is incomplete: modality affects not only attentional load but also *motivation*.

The tilt effect ($\chi^2 = 67.2$, $p < .001$) demonstrates that trial independence cannot be assumed in sequential motor tasks: a single miss increases the next-trial miss probability by 5$\times$.

Practical implications. Visual feedback may inadvertently create speed pressure in safety-critical interfaces; auditory feedback may better preserve accuracy.

Limitations. Effect sizes were small to medium ($d = 0.27$–0.32). Accuracy was near ceiling ($\sim$96%), limiting the range of detectable error differences. The linear SAT model explains only 3.2% of variance; the Pareto-like boundary pattern suggests a constraint envelope rather than a linear function. Device variability was controlled as a covariate in the regression but still introduced noise in the paired tests. Convenience sampling (all 18–24 university students) limits generalizability. The visual border glow is inherently more salient than a tone; our results speak to feedback "as deployed" rather than pure modality.

Lessons learned. Counterbalancing confirmed no order effect ($p = .960$). Gamification boosted engagement without biasing within-subject comparisons. The gold/green speed indicator, intended as neutral feedback, became a performance incentive, which is itself evidence that visual feedback drives speed-chasing behavior. Future work should manipulate feedback salience independently of modality to disentangle interference and motivational effects.

References

[1] C. D. Wickens, “Multiple resources and performance prediction,” *Theor. Issues Ergon. Sci.*, 3(2), 159–177, 2002.

[2] C. D. Wickens et al., *Engineering Psychology and Human Performance*, 4th ed., Routledge, 2015.

[3] J. Sweller, “Cognitive load during problem solving,” *Cognitive Science*, 12(2), 257–285, 1988.

[4] P. M. Fitts, “Information capacity of the human motor system,” *J. Exp. Psychol.*, 47(6), 381–391, 1954.

[5] I. S. MacKenzie, “Fitts’ law as a research and design tool in HCI,” *Hum.-Comput. Interact.*, 7(1), 91–139, 1992.

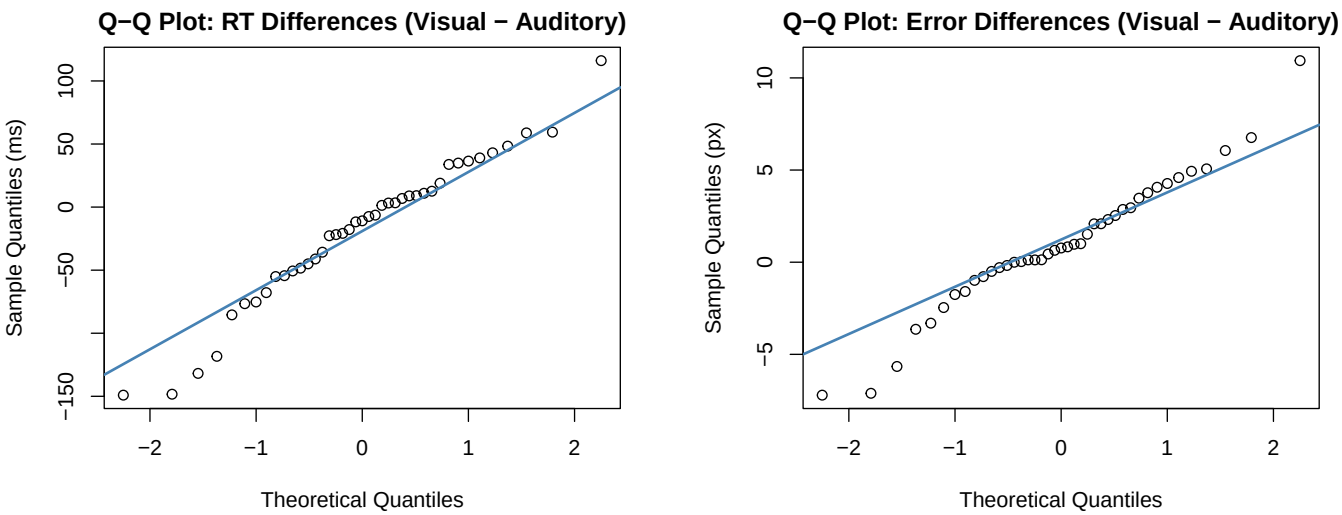
[6] R. P. Heitz, “The speed-accuracy tradeoff: history, physiology, methodology, and behavior,” *Front. Neurosci.*, 8, 150, 2014.

[7] M. Akamatsu et al., “Tactile, auditory, and visual feedback in a pointing task,” *Ergonomics*, 38(4), 816–827, 1995.

[8] R. Sigrist et al., “Augmented visual, auditory, haptic, and multimodal feedback in motor learning,” *Psychon. Bull. Rev.*, 20(1), 21–53, 2013.

A Assumption Checks

- RT differences: $W = 0.980$, $p = .304$: normal. Paired t-test appropriate.
- Error differences: $W = 0.420$, $p < .001$: non-normal. Wilcoxon used.



(a) RT differences:approximately normal. **(b)** Error differences:heavy tails, non-normal.

Figure 6: QQ plots for within-subject differences (visual − auditory).

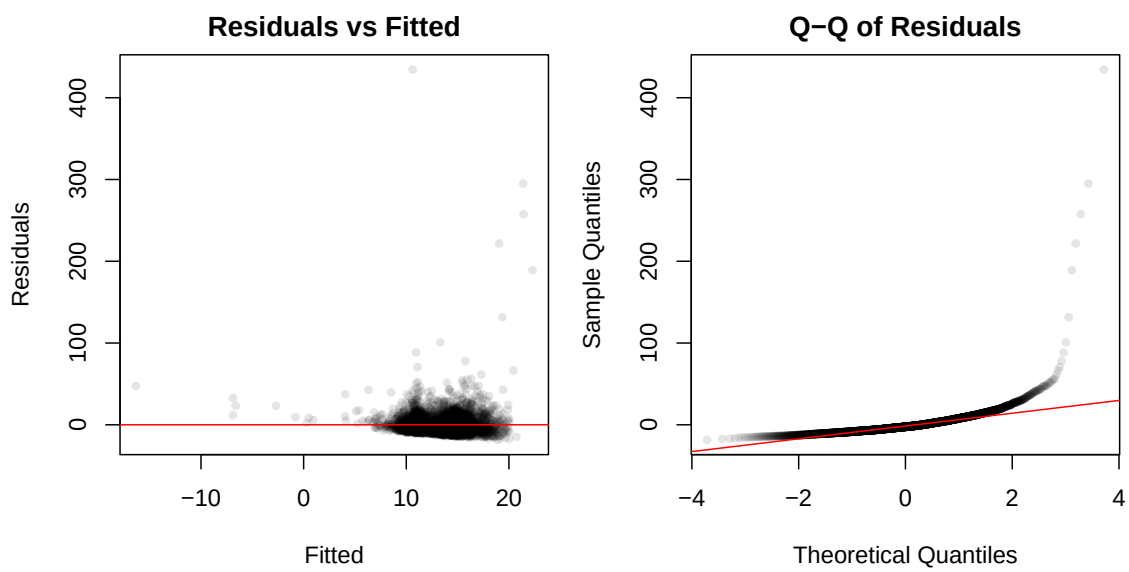
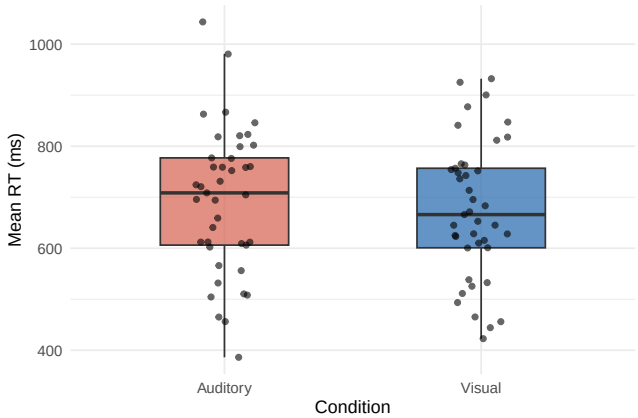
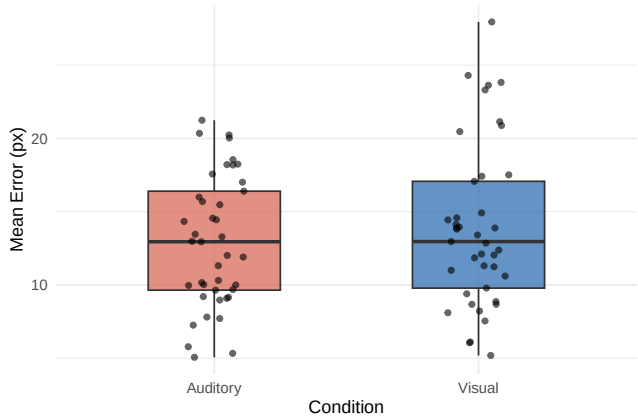


Figure 7: Residual diagnostics for the SAT interaction model. Funnel pattern expected with bounded error; heavy right tail from large misses.

B Additional Figures

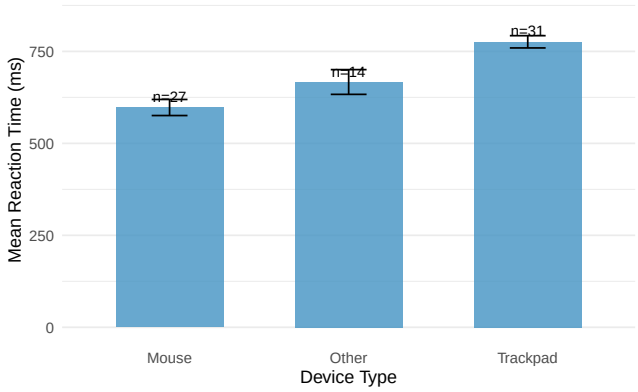


(a) RT by condition with individual points.

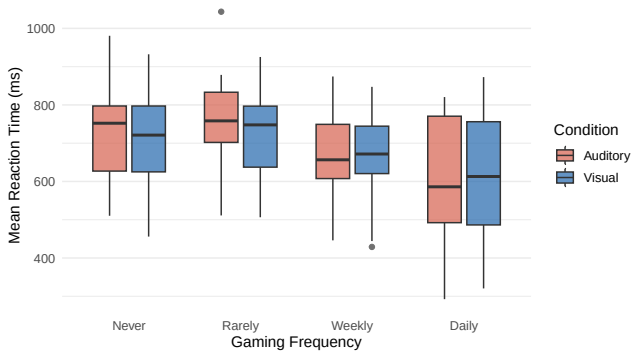


(b) Radial error by condition.

Figure 8: Boxplots of per-participant means by feedback condition.

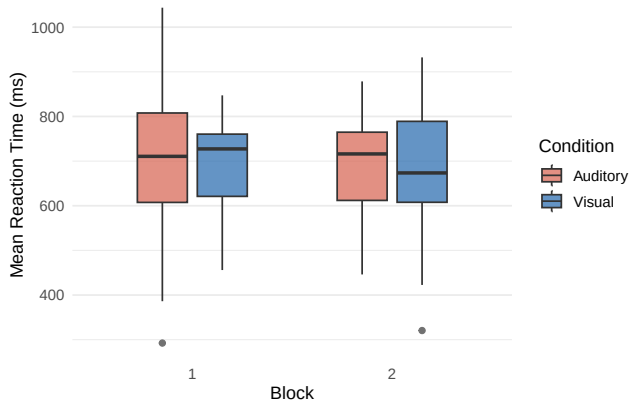


(a) RT by device type ($\pm$1 SE).

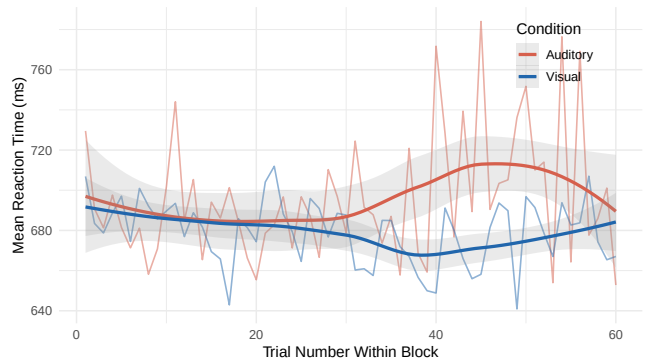


(b) RT by gaming frequency and condition.

Figure 9: Covariate effects on reaction time.

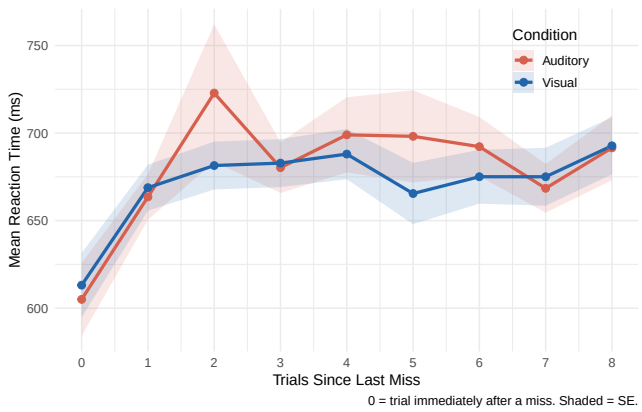


(a) RT by block ($p = .281$, $d = 0.13$).

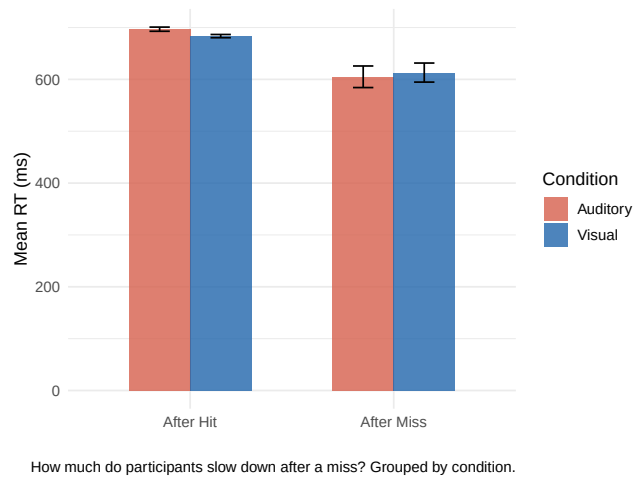


(b) RT by trial number within block.

Figure 10: Block and learning effects.



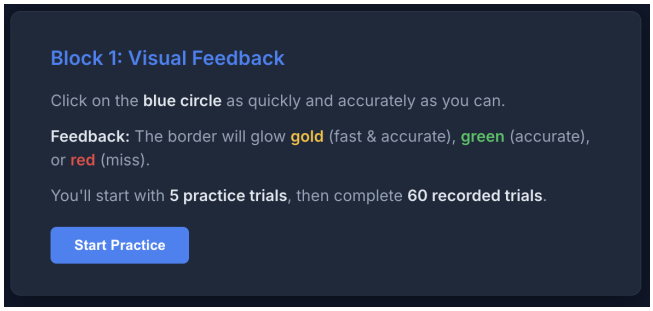
(a) RT trajectory after a miss. Visual participants slow down 65.5 ms vs. 17.4 ms for auditory.



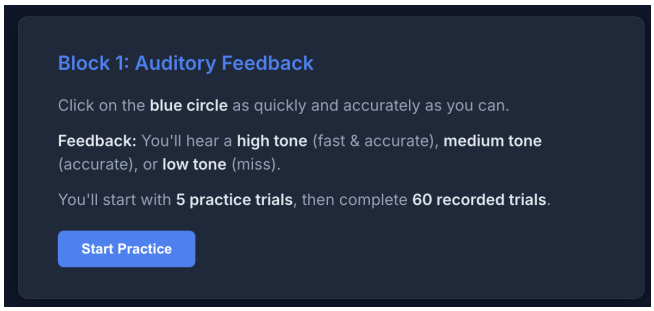
(b) Mean RT after hit vs. after miss, by condition.

Figure 11: Post-miss behavior: visual feedback makes failure more salient.

C Experiment Interface

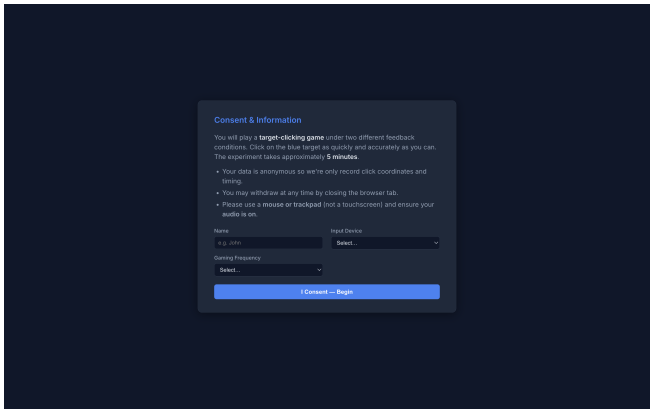


(a) Visual condition instructions.

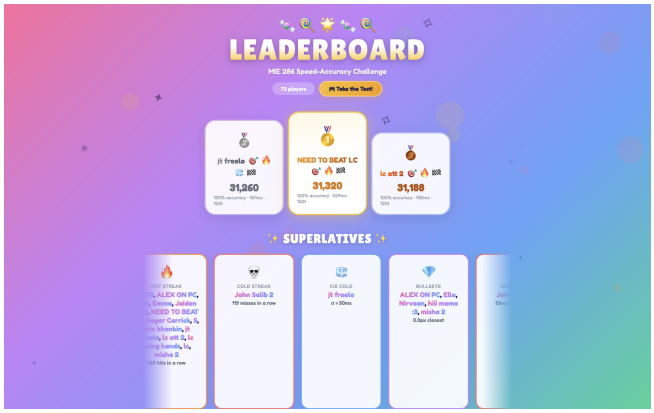


(b) Auditory condition instructions.

Figure 12: Instruction screens. Both describe the same task; only the feedback description differs.



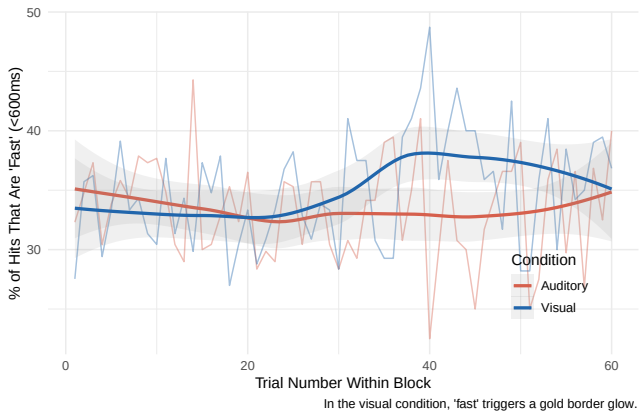
(a) Consent and demographics.



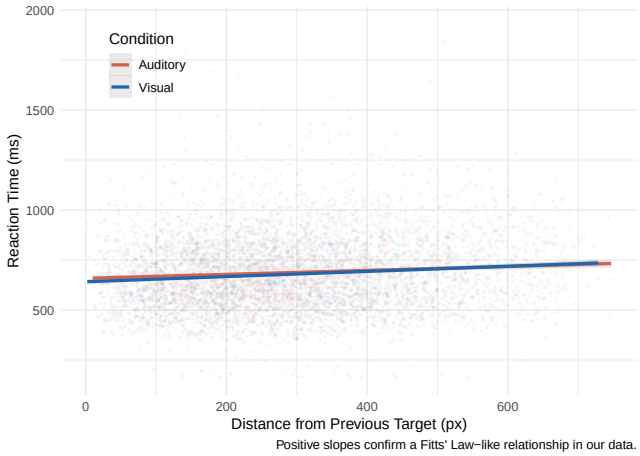
(b) Leaderboard with superlatives.

Figure 13: Data collection interface and leaderboard.

D Additional Analyses

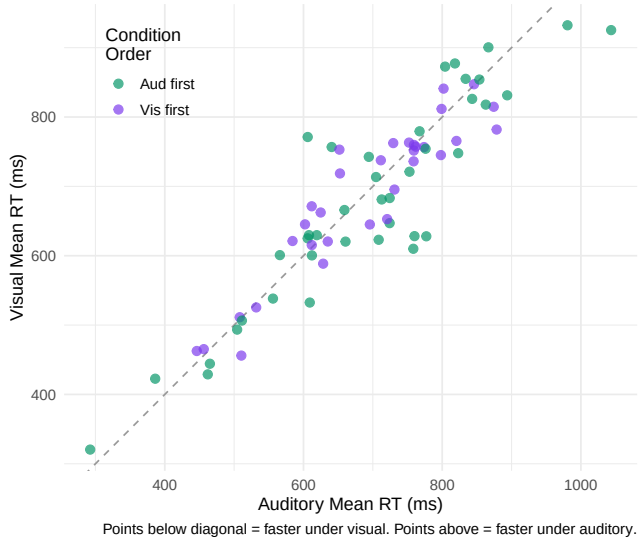


(a) Gold-chasing: % of fast ($<$600 ms) hits over trial number. Visual condition triggers gold glow for fast hits.

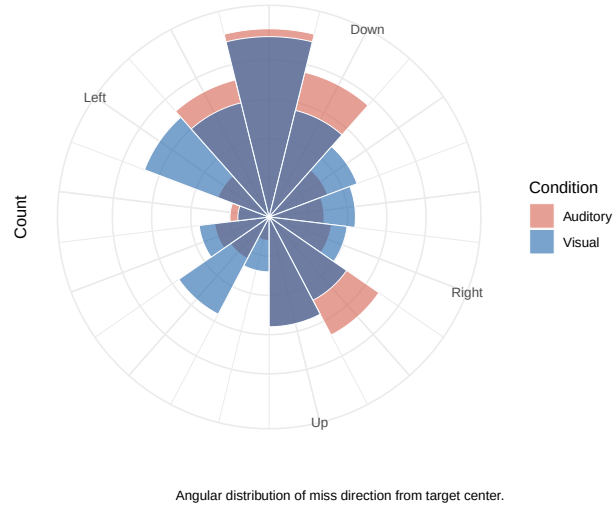


(b) Fitts' Law: inter-target distance predicts RT ($p < .001$). Visual slope is steeper (0.13 vs 0.10 ms/px).

Figure 14: Gold-chasing behavior and Fitts' Law relationships by condition.



(a) Per-participant: visual RT vs auditory RT. Below diagonal = faster under visual.



(b) Polar histogram of miss direction from target. Shows systematic rightward and downward bias.

Figure 15: Individual differences and spatial miss patterns.

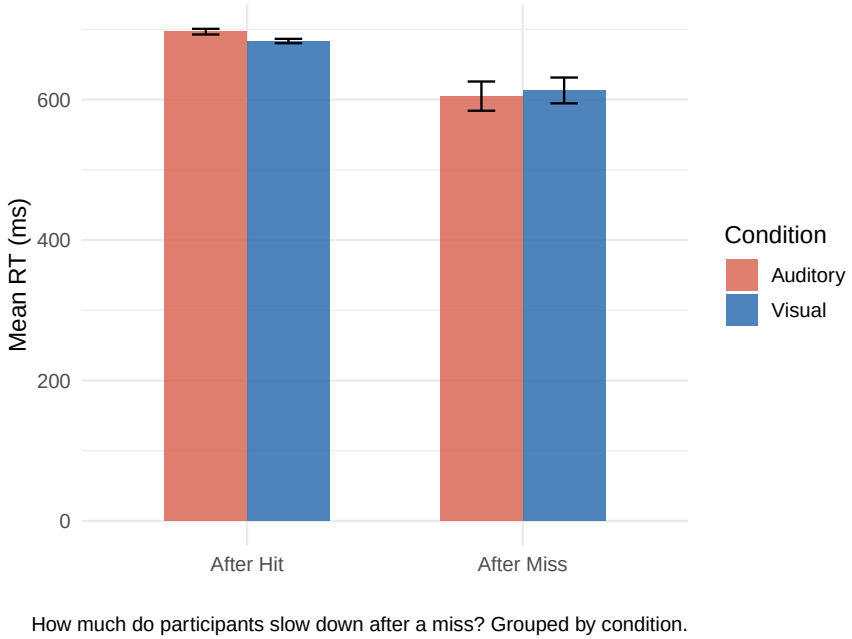


Figure 16: Mean RT after a hit vs. after a miss, by condition. Visual participants slow down 65.5 ms after a miss vs. 17.4 ms for auditory, suggesting visual feedback makes failure more salient.

E R Code

All analyses were conducted in R 4.5. The pipeline consists of six scripts run sequentially via `00_run_all.R`:

1. `01_load_clean.R` : Load all CSVs, exclude RT outliers, filter to 60-trial participants ($N = 41$), aggregate per-participant means.
2. `02_descriptives.R` : Summary tables by condition, device, gaming frequency.
3. `03_assumptions.R` : Shapiro–Wilk tests and QQ plots.
4. `04_inferential.R` : Paired t-test (H_1), Wilcoxon (H_2), multiple regression with interaction (H_3).
5. `05_plots.R` : All figures.
6. `06_secondary.R` : Tilt effect, order check, variance, block comparison, spatial bias.

Full code is in the `analysis/scripts/` directory submitted alongside this report.

F 30 vs. 60 Trial Comparison

To justify the shift from 30 to 60 trials per block, we compared the statistical results across both cohorts. Table **??** shows that the 30-trial participants ($N = 31$) produced no significant effects for any hypothesis, while the 60-trial participants ($N = 41$) produced significant results for all three. The 30-trial group added between-participant noise without contributing sufficient within-participant data for stable estimates.

Table 5: Comparison of statistical results by trial count.

	30 trials ($N = 31$)		**60 trials ($N = 41$)**	
	Statistic	p	Statistic	p
H_1: RT	$t = 0.21$	.837	$t = -2.05$	.047*
H_2: Error	$V = 203$	.811	$V = 580$	.026*
H_3: Interaction	$\beta_3 = 0.018$	.163	$\beta_3 = -0.011$	$< .001$***
Mean RT diff (ms)	$+2.0$		-18.5	
Mean error diff (px)	-2.5		$+1.0$	

The 30-trial cohort showed essentially zero RT difference between conditions ($+0.5$ ms) and an error difference in the *wrong direction* (-2.5 px). With only 30 trials per condition per participant, individual means were too noisy to detect the small within-subject effects. This motivated the design change to 60 trials, which provided sufficient statistical power ($d = 0.27$–0.32) to detect the feedback modality effect.